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Amrit Science Campus, Thamel, KATHMANDU

Institute of Science and Technology



A Project Report on

“Foodmine”

Submitted in partial fulfillment of the requirements for the course

Software Engineering (CSC 364)

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ABSTRACT

The Online Food Ordering System is a web-based application that allows users to browse, select, and order food online efficiently. The system aims to simplify the food ordering process by providing a user-friendly interface where customers can register, log in, and explore a variety of food options categorized based on type and availability. Each food item includes essential details such as price, preparation time, and images.

Users can search for their favorite meals, add them to a favorites list, and place orders by adding items to the cart. The system enables users to modify the quantity of items before proceeding to checkout. During checkout, customers are required to provide their name and address, with the additional feature of selecting their location via an interactive map for precise delivery details.

The payment system is integrated with secure methods, including e-banking and Khalti, ensuring a seamless transaction process. An admin panel allows restaurant owners or administrators to manage orders, view customer details, and update the food menu as necessary.

The project enhances the efficiency of food ordering services by reducing manual intervention, minimizing errors, and improving customer satisfaction. It also offers a more streamlined order management system for restaurant owners. The future scope of the project includes expanding the platform to support a mobile application, incorporating additional payment gateways, and adding advanced features like food reviews, discount coupons, and real-time order tracking.

Keywords: *Online Food Ordering, E-Commerce, Food Delivery System, User Authentication, Shopping Cart, Online Payment, Khalti Integration, Order Management System, Web-Based Application.*

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CHAPTER 1:

INTRODUCTION

1.1 Background

In today's digital era, online food ordering has become a convenient and efficient way for customers to order meals from their favorite restaurants. Traditional methods of food ordering, such as phone calls or physical visits, can be time-consuming and prone to errors. To address these challenges, Foodmine, an E-Commerce Food Ordering System, has been developed to provide a seamless and user-friendly platform for food lovers.

Foodmine allows users to browse, search, and order food items online with essential details such as price, cooking time, and images. It includes features like user authentication, a dynamic menu, a favorites option, a shopping cart, and a checkout process with secure payment options. Customers can select their delivery address using a map integration feature, ensuring accurate and hassle-free deliveries.

1.2 Problem Statements

The traditional food ordering process involves several challenges that can lead to inconvenience for both customers and restaurant owners. Foodmine aims to address these issues by providing a streamlined and efficient online food ordering system. The key problems that this project solves are:

- **Time-consuming ordering process:** Customers often spend time waiting in restaurants or making phone calls to place orders, which can be inefficient.
- **Limited accessibility:** Traditional ordering methods may not be available 24/7, restricting customers from placing orders at their convenience.
- **Lack of menu visibility:** Customers may not have access to a restaurant's complete menu, pricing, or preparation time before placing an order.
- **Manual order management issues:** Restaurants handling phone orders are prone to miscommunication, leading to incorrect orders and customer dissatisfaction.
- **Payment inconvenience:** Many traditional restaurants rely on cash payments, which can be inconvenient for customers who prefer online transactions.
- **Limited admin control:** Restaurant owners may struggle with order management, tracking customer details, and updating food items efficiently.

1.3 Objectives

- To develop a fully-functional E-Commerce Food Ordering System.

1.4 Application

- Enables users to order food online from various restaurants easily.
- Provides a secure and user-friendly ordering and payment system.
- Allows users to search for food, view details, and add items to the cart.
- Offers multiple payment options, including e-banking and Khalti.
- Allows users to save favorite food items for quick access.
- Provides a seamless checkout process with address selection via a map.
- Helps restaurant owners manage orders efficiently through an admin panel.

1.5 Features

Prime features of this project are:

1. **User Authentication (Login & Signup):** Secure user registration and login process. It prevents unauthorized access and ensures user data security.
2. **Home Page with Food Listings:** Displays a variety of food items with images, prices, and preparation time. Well-organized categories to help users explore different food options.
3. **Search Functionality:** Users can search for specific food items quickly. It provides real-time search results based on user preferences.
4. **Food Details Page:** Shows detailed information about each food item, including price, description, and estimated cooking time.
5. **Favorites Option:** Users can mark food items as favorites for quick access later. It makes it easier to reorder frequently ordered items.
6. **Add to Cart & Cart Management:** Users can add multiple food items to the shopping cart. Provide option to increase or decrease the quantity of each item. Users can remove unwanted items before checkout.
7. **Checkout Process:** Displays the customer's name and address before placing the order. Integrated map feature to select an accurate delivery location.
8. **Secure Online Payment:** Supports e-banking and Khalti for hassle-free transactions. Ensures secure and quick payments without the need for cash.
9. **Order Tracking & Confirmation:** Users receive order confirmation after a successful transaction. Estimated cooking and delivery time is displayed.

10. Admin Panel for Order & Menu Management: Admins can view and manage orders placed by customers. Allows restaurant owners to add, update, or remove food items from the menu.

1.6 System requirements

1.6.1 Software requirement

For the program to run, the set of following software to be prepared with the following specifications:

Frontend:

- **Framework:** Angular (for building the user interface)
- **Languages:** TypeScript, HTML, CSS
- **Package Manager:** Node Package Manager (NPM)
- **Additional Libraries:** Bootstrap, Angular Material

Backend:

- **Server-Side Framework:** Node.js with Express.js
- **Database:** MongoDB (NoSQL database for storing user, food, and order data)
- **Authentication:** JSON Web Token (JWT) for secure login authentication
- **Payment Integration:** Khalti API, e-banking API

Development & Hosting Tools:

- **Code Editor:** Visual Studio Code
- **Version Control:** Git & GitHub
- **Database Management Tool:** MongoDB Compass

1.6.2 Hardware requirements

For this program to be run, the computer hardware must be prepared with the following specifications:

- **Processor:** Intel i5 or higher
- **RAM:** 8 GB or more
- **Storage:** 50 GB of free space or higher

1.7 Feasibility analysis:

1.7.1 Economic feasibility

Foodmine is cost-effective, using open-source technologies like Angular, Node.js, Express, and MongoDB, which reduce development expenses. The platform offers revenue opportunities through restaurant subscriptions, commissions, and advertisements, ensuring profitability.

1.7.2 Technical feasibility

Foodmine uses a reliable tech stack (Angular, Node.js, Express, and MongoDB) that ensures scalability, fast processing, and security. The system can handle multiple users and transactions efficiently, with secure payment integrations and real-time order management.

1.7.3 Operational feasibility

The system is operationally efficient, offering an easy-to-use interface for customers and an intuitive admin panel for restaurant owners. Features like order tracking, map-based delivery, and secure payments streamline daily operations, ensuring smooth management for both users and businesses.

CHAPTER 2: LITERATURE REVIEW

The food industry in Nepal is experiencing a digital shift, with an increasing number of consumers turning to online platforms for food ordering. According to **Adhikari & Sharma (2021)**, the adoption of **online food delivery services** like **Foodmandu** and **Daraz Food** in Nepal is growing rapidly due to the increasing use of smartphones and the convenience offered by such platforms. This trend highlights the growing demand for online food ordering systems in Nepal, creating an opportunity for new platforms like **Foodmine**. [1]

A study by Khadka & Paudel (2020) explored consumer behavior in Nepal's food ordering market. The research shows that Nepali consumers are becoming increasingly accustomed to ordering food online, primarily due to convenience, better deals, and more diverse food options. Foodmine, by offering an intuitive, user-friendly interface, can cater to this growing trend and meet the demand for easy and efficient food ordering experiences. [2]

According to Shrestha & Bhandari (2021), one of the major challenges in Nepal's e-commerce sector is the lack of robust digital payment systems. However, with the rise of payment services like Khalti and e-banking, the situation is improving. The research suggests that platforms like Foodmine, integrating secure local payment gateways such as Khalti and e-banking, will gain the trust of consumers and provide a safe environment for online transactions in Nepal. [3]

A study by Koirala & Adhikari (2020) highlights the increasing use of mobile apps for food ordering in urban areas of Nepal. With a large portion of Nepal's population relying on smartphones, mobile-based platforms offer greater accessibility. Foodmine's focus on mobile app integration will cater to a growing segment of mobile users, providing a convenient food ordering solution for Nepali consumers. [4]

The research by Ghimire & Thapa (2022) discusses the role of real-time tracking and GPS navigation in improving delivery efficiency in Nepal. With the challenges posed by Nepal's complex terrain and traffic, integrating real-time order tracking into Foodmine will help enhance the delivery experience by ensuring timely and accurate deliveries, which is crucial for customer satisfaction in Nepal's food delivery market. [5]

CHAPTER 3: METHODOLOGY

The development of **Foodmine** follows a **semi-waterfall approach**, involving the following phases:

- **Requirement Analysis:** Identifying the needs of users and restaurants to define essential features like authentication, search, cart management, and secure payment.
- **Design:** Creating wireframes, mockups, and prototypes to structure the user interface (UI) and user experience (UX) for seamless navigation.
- **Development:** Implementing the frontend using Angular, the backend with Node.js and Express, and MongoDB for database management to ensure scalability and efficiency.
- **Testing:** Conducting manual and automated testing to verify functionality, security, and performance, ensuring a smooth user experience.
- **Deployment:** Deploying Foodmine on a local host.

3.1 Block diagram

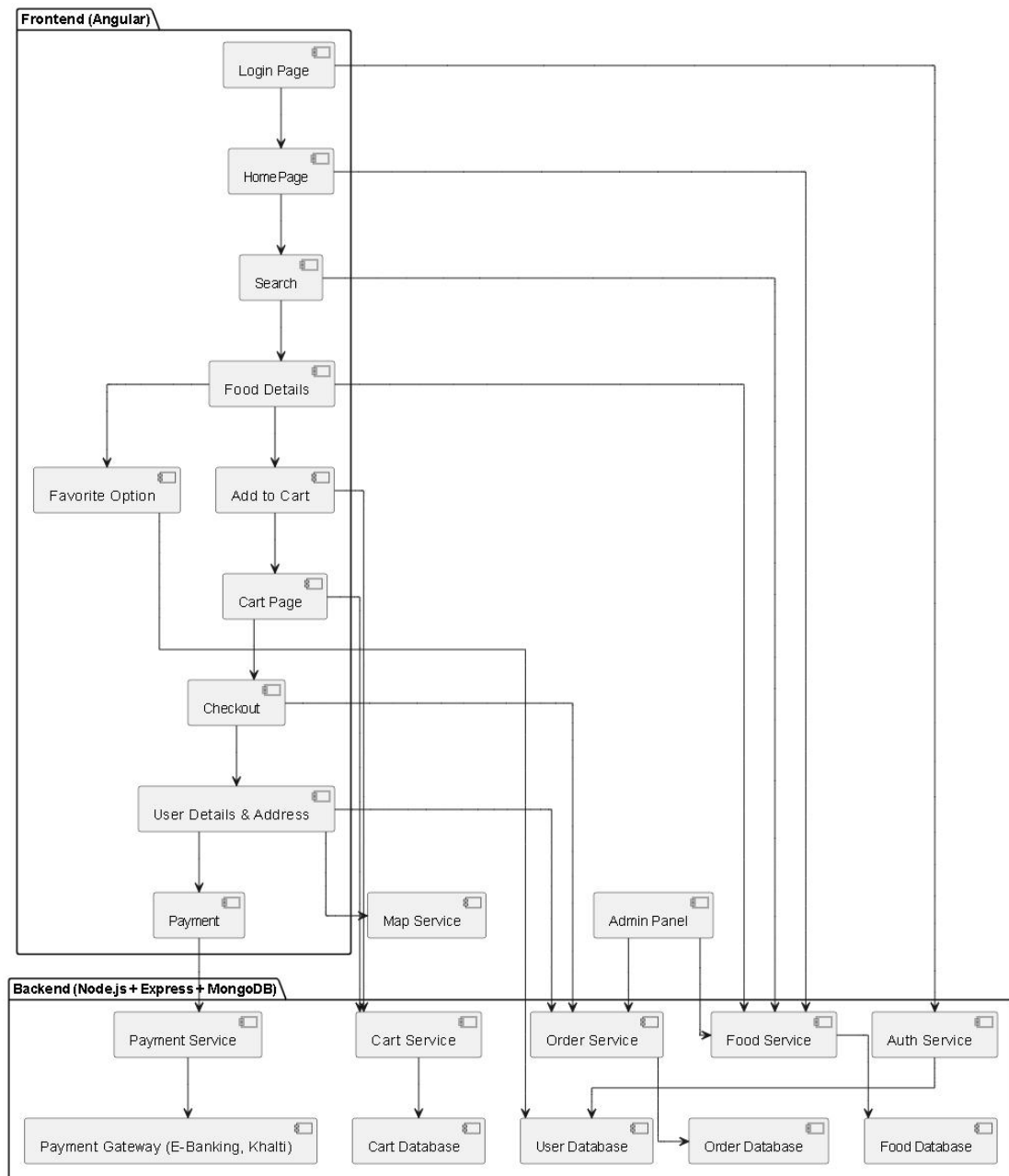


Figure 1:Block Diagram

3.2 Use Case diagram

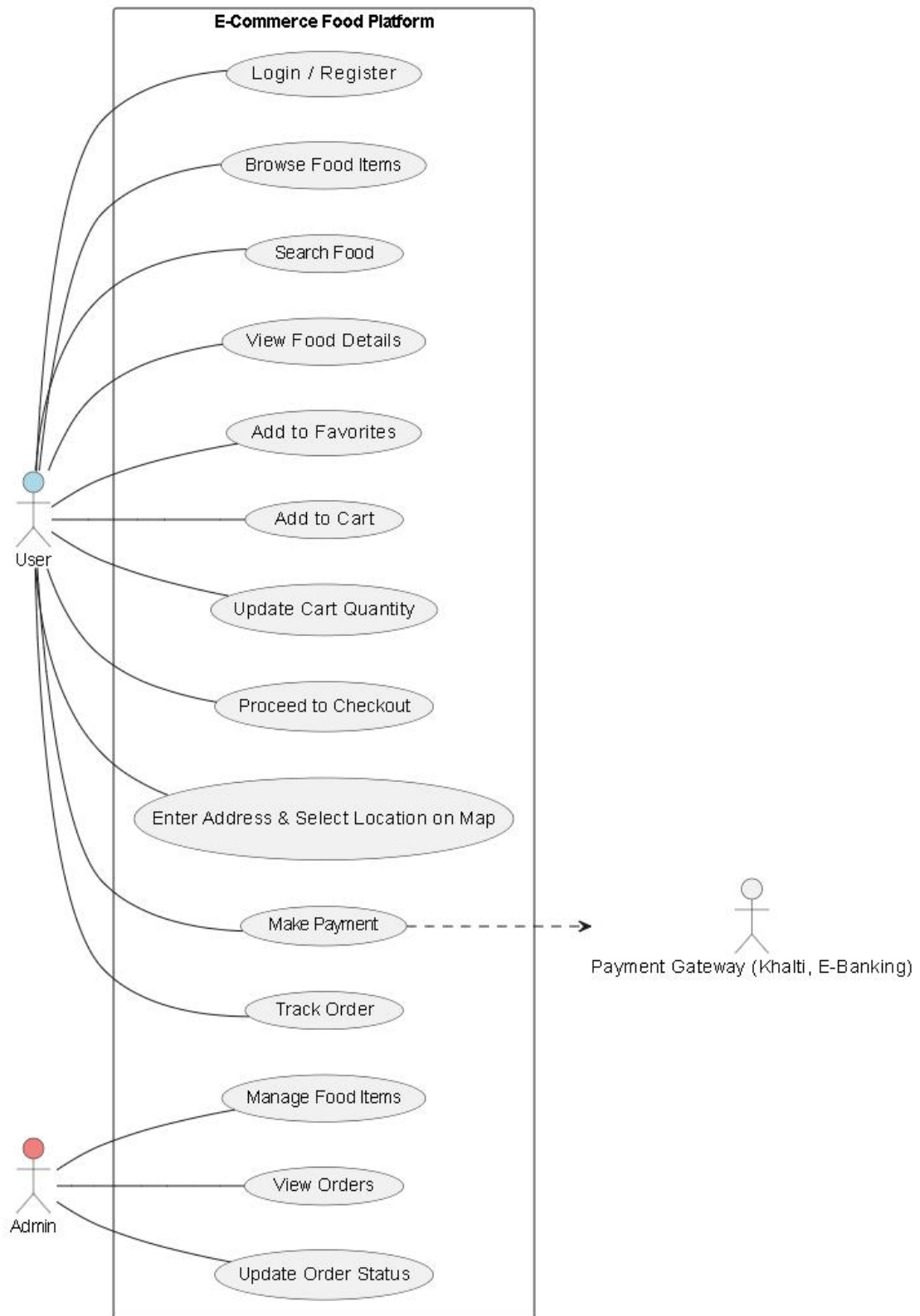


Figure 2: Use Case Diagram

3.4 Gantt Chart

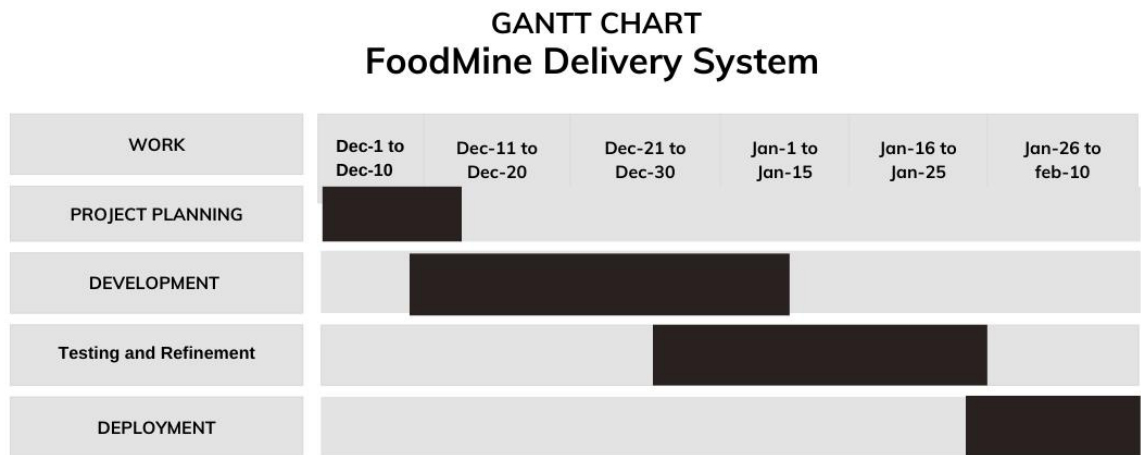


Figure 3: Gantt Chart

CHAPTER 4: RESULTS AND OUTCOME

4.1 Outcome

The screenshot shows the Login page of the FoodMine! application. At the top, a red header bar contains the text "FoodMine!" on the left and "login" and "Cart" on the right. The main content area is white and features a "Login" heading. Below the heading are two input fields: "Email" and "Password", each with a placeholder text of the same name. A red "Login" button is positioned below the input fields. At the bottom of the form, there is a link that says "New user? Register here".

Figure 4: Login

The screenshot shows the Register page of the FoodMine! application. At the top, a red header bar contains the text "FoodMine!" on the left and "login" and "Cart" on the right. The main content area is white and features a "Register" heading. Below the heading are five input fields: "Name", "Email", "Password", "Confirm Password", and "Address", each with a placeholder text of the same name. A red "Register" button is positioned below the input fields. At the bottom of the form, there is a link that says "Already a user? Login Here".

Figure 5: Register Page

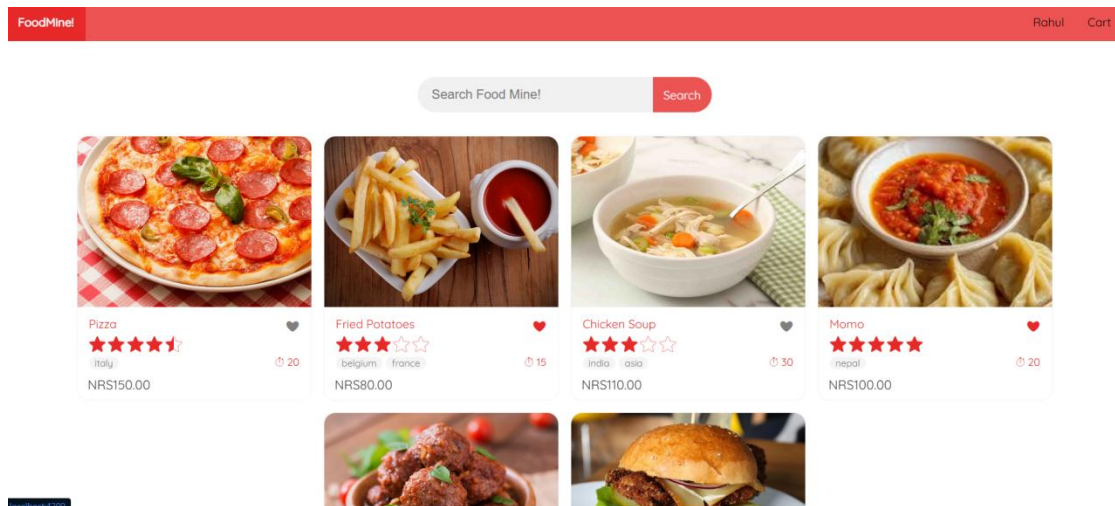


Figure 6: Home Page

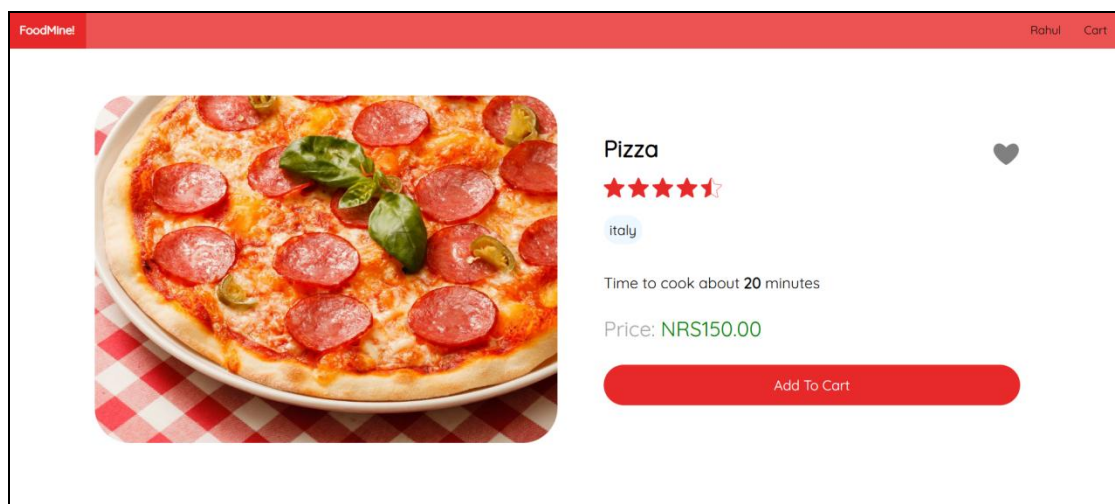


Figure 7: Food Detail Page

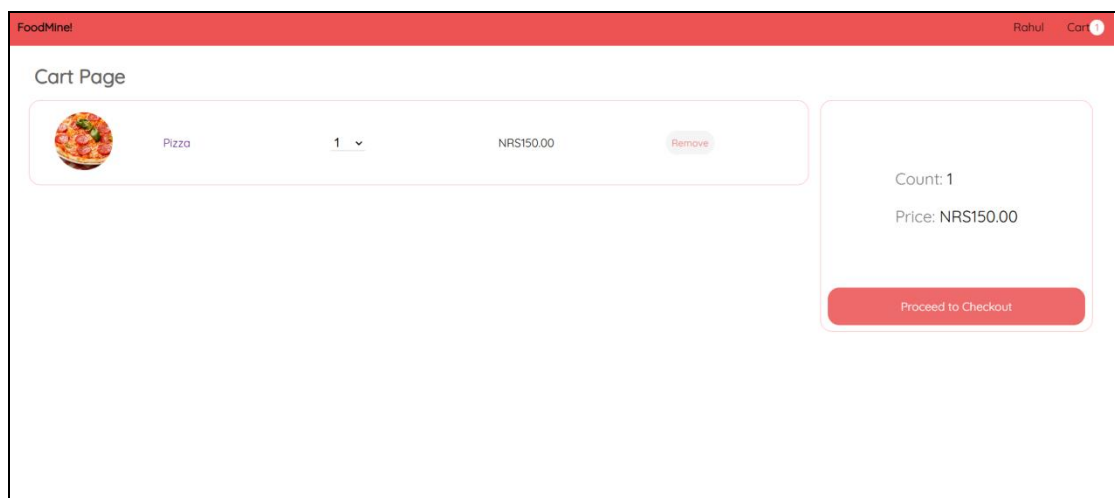


Figure 8: Cart Page

FoodMine!

RahulCart1

Order Form

Name

Rahul

Address

Yaksho

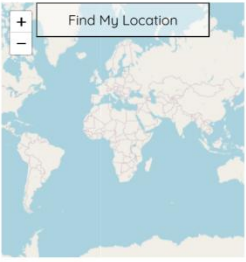
Order Items:

 Pizza NRS150.00 1 NRS150.00

Total : NRS150.00

Choose Your Address On Map

+
-
Find My Location



Go To Payment

Figure 9: User Detail Page

FoodMine!

RahulCart1

Order Summary

Name: Rahul

Address: Yaksho

Order Items:

 Pizza NRS150.00 1 NRS150.00

Total : NRS150.00

 Pay with Khalti

Figure 10: Payment details Page

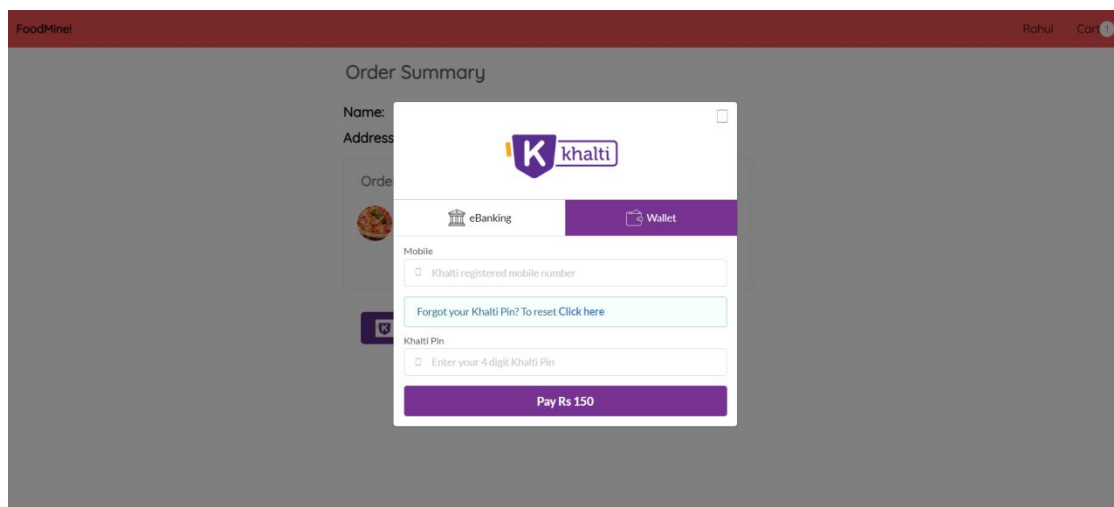


Figure 10: Payment Page

CHAPTER 5: CONCLUSION

The Foodmine project is an innovative e-commerce platform for food ordering designed to enhance the convenience of both customers and restaurant owners in Nepal. By leveraging modern technologies such as Angular, Node.js, Express, and MongoDB, the system ensures a fast, secure, and scalable experience. Features like user authentication, search functionality, food details, cart management, real-time order tracking, and secure payments via Khalti and e-banking make it a comprehensive solution for online food delivery.

The feasibility analysis confirms that Foodmine is economically viable, technically robust, and operationally efficient. The literature review highlights the growing demand for online food ordering platforms in Nepal, the importance of mobile technology, and the need for secure payment systems. By addressing these factors, Foodmine has the potential to revolutionize the food delivery industry in Nepal, providing a reliable experience for both customers and businesses.

CHAPTER 6: FUTURE ENHANCEMENT

Mobile Application Development – Develop a dedicated Android and iOS app for a smoother user experience, as most Nepali users prefer mobile-based ordering.

AI-Based Food Recommendations – Implement machine learning algorithms to suggest food items based on user preferences, order history, and popular trends.

Multi-Vendor Support – Expand the platform to allow multiple restaurants and home-based food businesses to register and manage their own stores.

Loyalty and Discount System – Introduce a reward points system and personalized discounts to encourage customer retention and repeat orders.

Cash on Delivery (COD) Option – Add the COD payment option, as many Nepali users still prefer cash transactions.

Integration with Food Delivery Services – Partner with local delivery services for faster and more efficient food delivery across different cities in Nepal.

Customer Feedback & Reviews – Implement a detailed rating and review system to enhance service quality and customer trust.

Subscription-Based Meal Plans – Introduce a meal subscription feature where users can pre-order meals for a week or month, benefiting office workers and students.

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Certificate of Project Approval

It is hereby approved that this project report on “**Foodmine – E-commerce Platform for Food Ordering**” is a creditable study of the project and has been presented in a satisfactory manner by **Saharsh Gurung, Nirman Nepali, Raman Pun Magar and Rahul Pajiyar** to warrant its acceptance as a prerequisite to obtaining the degree for which it has been submitted.

In granting this approval, the undersigned acknowledge that they are not necessarily endorsing any conclusions or opinions in the report but are approving it for the purpose it was intended.

(Internal Examiner)

(External Examiner)

(Head of Department)